VOLUME theorems_v2

Theorems v2.0 (T11–T18)

Hilbert Polya Bridge · v2.0 «The Operator That Almost Exists»

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Обозначения / Notation

γ_n — zeta zero ordinates; Λ(n) — von Mangoldt function; Φ_p — prime flow; Q(a), K(x) — sides of the explicit formula (see docs/trace_formula.md v1); ε — staggered structure; L — lattice size.

T11. Heat-trace uniqueness

Theorem T11.

If Tr e^{−xH} = $(1/\pi)\int_0^\infty Q(a)\sin(ax)da$ for all x > 0, where Q(a) = Σ_{γ>0} $2a/(a^2+\gamma^2)$, then spec H = {±γ_n} with multiplicity. Proof: both sides are Laplace–Stieltjes transforms; uniqueness of the measure; the per-zero identity.

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: доказано / proven.

T12. Co-location classification

Theorem T12.

Phase systems with Φ_pq ≡ Φ_p + Φ_q (mod 2π) are homomorphisms ⟨p⟩ → ℝ/2πℤ; normal form Φ_p = 2π{β ln p}. {0, π}-valued systems are gauge-trivial in the overlay channel (ψ → (−1)^y ψ).

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: доказано+измерено / proven+measured (4.4e-16; 8.4e-15).

T13. α=1/3 counting law

Theorem T13.

$N_0 = 2L - 6\cdot[L \equiv 0 \pmod{12}]$ for L ≡ 0 (mod 6); else $N_0 = 0$. Index Σ sign(ε) = $-2\cdot[L \equiv 0 \pmod{12}]$. Decisive checks: L=36→66, L=48→90.

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: измерено (закон) / measured (law).

T14. Band T=5000

Theorem T14.

RH verified in the band T ≤ 5000: N_online = N_strip = 4520, Ν = 4520.08, S_max = 0.91 ≤ 2, v1 anchors reproduced.

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: измерено / measured.

T15. Blind demarcation

Theorem T15.

20 gates from 5 new families (outside the v1 handle space) — all in the null zone ($|r_{H7}| \le 0.040$) with PC0' = 1.000.

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: измерено / measured.

T16. Attractor taxonomy

Theorem T16.

Scan of 121 configurations: 97 GUE / 18 integrable ($N_v=0$) / 6 intermediate ($N_v\le 8$, $q=0.5$, $\langle r\rangle=0.47$-0.53); no rank outliers.

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: измерено / measured.

T17. Twisted trace formula

Theorem T17.

$K_\chi(x) = \Sigma e^{-x\gamma_\chi} = (1/\pi)\int Q_\chi(a)\sin(ax)da$, $Q_\chi = \Lambda'/\Lambda(\tfrac{1}{2}+a,\chi) - B_\chi - C_0^\chi$, $C_0^\chi = \Lambda'/\Lambda(\tfrac{1}{2}) - B_\chi$; carries $\chi(n)\Lambda(n)/\sqrt{n}$. Verified to $|\Delta K| \le 9.6e\text{-}6$ ($\chi 5$), $6.1e\text{-}6$ ($\chi 7$).

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: доказано+верифицировано / proven+verified.

T18. Quantum echo of locality

Theorem T18.

The gate's Fermi-sea fermionic entanglement is suppressed by locality ($S_A \approx$ 12-16 nats vs 160-220 Haar baseline, $z > 100$); the MI profile carries no zero signature (-0.018 vs -0.121 random).

Proof. *See monograph v2.0, the corresponding chapter; full derivations in computations/C*/README and docs/v2/.*

Status: измерено / measured.